

Customer Segmentation with Clustering

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1. Introduction / Overview of the Problem

In highly competitive e-commerce environments, understanding customer behaviour is essential for designing effective marketing strategies and improving customer retention. Rather than treating all customers as a homogeneous group, organisations can benefit from identifying distinct customer segments based on purchasing behaviour and demographic characteristics.

This project focuses on customer segmentation using clustering techniques to support targeted marketing and strategic decision-making. The dataset used is a real-world, transnational e-commerce dataset containing customers from five continents (Oceania, North America, Europe, Africa, and Asia) and 47 countries (SAS, 2024). Given the scale and diversity of the data, manual segmentation is impractical, making unsupervised machine learning an appropriate solution.

The goal of this project is to group customers into meaningful and distinct segments without predefined labels, allowing the business to tailor marketing efforts, optimise customer engagement strategies, and allocate resources more effectively.

2. Overview of the Approach

The project followed a structured and systematic data science workflow:

1. **Data Cleaning and Preprocessing:** The dataset was cleaned by handling missing values, correcting data types, and removing duplicate records to ensure data quality and consistency.

2. **Feature Engineering:** To enhance the clustering process, five new features were engineered to better capture customer behaviour and characteristics. These engineered features were stored in a separate dataframe specifically prepared for clustering analysis.

3. **Exploratory Data Analysis (EDA):** EDA was performed on both the original dataset and the feature-engineered dataset to examine distributions, identify patterns, and understand relationships between variables.

4. **Clustering Model Selection:** Unsupervised clustering was performed using K-Means, with careful consideration given to selecting the optimal number of clusters.

5. **Dimensionality Reduction and Visualisation:** Dimensionality reduction techniques (PCA and t-SNE) were applied to visualise the clustering results and assess cluster separation.

This end-to-end approach ensured that the clustering results were both statistically sound and interpretable from a business perspective.

3. Customer Segmentation Methodology

Before applying K-Means clustering, it was necessary to determine the most appropriate value of k . Three complementary methods were used:

- **Elbow Method:** Examined the within-cluster sum of squares to identify diminishing returns as k increases.

- **Silhouette Analysis:** Measured how well each observation fit within its assigned cluster relative to other clusters.

- Agglomerative Hierarchical Clustering (Dendrogram): Provided a hierarchical view of cluster merging patterns.

While both the elbow method and dendrogram suggested ambiguity between five and six clusters, silhouette analysis clearly indicated that five clusters achieved the best separation, with the highest silhouette score. Consistency across these methods supported the selection of $k = 5$ as the optimal solution.

4. Key Results: Figures, Tables & Interpretation of the Clustering Results

The five clusters (labelled 0–4) exhibit subtle but meaningful differences across the engineered features, indicating variations in customer behaviour and demographics rather than extreme segmentation.

- Cluster 0: Customers in Cluster 0 show a slightly higher purchase frequency compared to the other clusters. Their customer lifetime value (CLV) is also marginally higher, although overall CLV is relatively consistent across all clusters. In terms of age, this cluster is similar to Cluster 1, indicating a mid-range age profile. This segment appears to represent moderately active customers with steady engagement.

- Cluster 1: Cluster 1 is characterised by a slightly higher recency in days, suggesting that these customers have, on average, taken longer since their most recent purchase compared to most other clusters (excluding Cluster 4). This cluster also shows a slightly higher average unit cost, indicating a tendency to purchase marginally higher-priced items. The age distribution is similar to Cluster 0, placing this segment in a comparable demographic group.

- Cluster 2: Customers in Cluster 2 exhibit higher age values, distinguishing them from younger segments such as Cluster 3. Behavioural features such as frequency, recency, CLV, and average unit cost are broadly similar to the overall dataset averages, suggesting that age is the primary differentiating characteristic for this cluster.

- Cluster 3: Cluster 3 stands out as the youngest customer segment, with noticeably lower age values than the other clusters. Despite this demographic difference, behavioural metrics such as frequency, recency, CLV, and average unit cost remain broadly aligned with the rest of the clusters, indicating that younger customers in this dataset behave similarly to older customers in terms of purchasing patterns.

- Cluster 4: Cluster 4 is primarily distinguished by a very high recency in days, indicating that these customers have not made purchases for a significantly longer period. This suggests lower recent engagement compared to all other clusters. Additionally, customers in this cluster tend to be older, similar to Cluster 2. Other behavioural features, including frequency, CLV, and average unit cost, do not differ substantially from the remaining clusters.

Overall, the clustering results indicate that:

- Recency and age are the most influential features in differentiating customer segments.
- CLV and average unit cost remain relatively stable across clusters, suggesting similar spending levels.
- The segmentation captures nuanced behavioural and demographic differences rather than sharply distinct customer groups, which is typical for real-world transactional e-commerce data.

5. Visualisation of Customer Segmentation

To visualise the high-dimensional clustering results, two dimensionality reduction techniques were applied:

- Principal Component Analysis (PCA): PCA provided a rough 2D projection of the clusters but showed limited separation. This is likely due to PCA's reliance on linear transformations, which may not fully capture complex, non-linear relationships in customer behaviour data.

- t-Distributed Stochastic Neighbor Embedding (t-SNE): t-SNE produced a significantly clearer visualisation, with all five clusters distinctly separated in two-dimensional space. This suggests that the underlying structure of the data is non-linear, making t-SNE a more suitable tool for visual interpretation in this context.

6. Conclusion

This project successfully implemented a customer segmentation framework using unsupervised K-Means clustering on a large, real-world e-commerce dataset. By combining preprocessing, feature engineering, multiple cluster validation techniques, and visualisation methods, five meaningful customer segments were identified.

The use of t-SNE provided clear visual confirmation of cluster separation, while boxplots enabled detailed comparison of customer characteristics across segments. These insights can support targeted marketing strategies, improved customer engagement, and more efficient allocation of business resources.